

Methodological quality and reporting completeness of machine learning models for mental health prediction: a systematic review and PROBAST+AI / TRIPOD+AI appraisal

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Protocol registered on PROSPERO: [TODO: CRD number]

Abstract

Background. Machine learning (ML) models that predict, detect or classify mental health conditions are published at an accelerating rate, and descriptive reviews of the field already exist. Whether these models are developed and reported to the standards now expected of clinical prediction models has not been assessed across populations, disorders and data modalities. **Methods.** We searched PubMed, Scopus, Web of Science and IEEE Xplore (1 January 2018 to [TODO: search date]) for studies developing or validating a supervised ML model predicting a mental health outcome in humans, with at least one performance metric. Two reviewers screened, extracted and appraised every study with PROBAST+AI (risk of bias and applicability) and the TRIPOD+AI checklist (reporting completeness). Pre-specified secondary outcomes were the proportions of studies with external validation, calibration assessment, discrimination reported as AUROC rather than accuracy alone, leakage-prone evaluation design, events per variable (EPV) relative to current guidance, and code or data availability. **Results.** [TODO: n records; n included; PROBAST+AI domain distribution; TRIPOD+AI adherence; secondary-outcome proportions with 95% CIs; stratification by modality.] **Conclusions.** [TODO: one sentence on prevalence of the deficiencies]. A companion empirical study on three public actigraphy cohorts quantifies what these deficiencies cost: record-wise cross-validation inflated AUROC by 0.04 to 0.17, external validation reduced AUROC by up to 0.21 with calibration slopes collapsing to 0.3, and no published model on those cohorts reports calibration at all.

1 Introduction

Prediction models based on machine learning are proposed for nearly every mental health condition and data source: electronic health records, questionnaires, wearable actigraphy, speech, neuroimaging and social media text. Descriptive reviews catalogue the algorithms and datasets in use [1, 2]; the most recent screened 3,320 records and organised the field by method and data type [1]. What none of them asks is whether the models are trustworthy: whether they were evaluated on participants unseen in training, whether their probabilities are calibrated, whether their sample sizes support the number of candidate predictors, and whether the reports contain what a reader needs to reproduce or appraise them.

For clinical prediction models generally, that question has been asked and the answer is discouraging. Meehan and colleagues appraised 308 psychiatric prediction models and found that 22.1% tested calibration and 11.4% performed internal validation to an adequate standard [3]; external validation

is rare and generalisability poorly characterised [4]. Those appraisals concentrate on regression-based clinical models. The ML literature that now dominates mental-health prediction, and in particular the sensor-, speech- and text-based studies, has been appraised only in narrow slices: adolescent EHR crisis prediction [5], psychosis prediction in at-risk individuals [6], ADHD [7], and social-media detection [8]. Two developments make a field-wide appraisal both feasible and overdue: TRIPOD+AI [9] and PROBAST+AI [10] extend the reporting and risk-of-bias instruments explicitly to ML models, adding items on data leakage, class imbalance, fairness, and ML-specific sample-size expectations.

Two evaluation practices are specific enough to ML on repeated-measures data to deserve their own outcome. First, *record-wise* splitting, in which windows or sessions from one participant appear on both sides of a train/test split, lets a flexible model recognise the participant rather than the condition; the resulting optimism is well documented in wearable and neuroimaging research [11, 12] but its prevalence in mental-health ML has not been counted. Second, resampling or feature selection applied before splitting leaks test information into training. Both are invisible in a results table and both are what turns a 0.75 AUROC into a 0.95.

We therefore asked: how methodologically sound and completely reported are ML models for mental health prediction, across all populations, disorders and modalities? We pre-specified the proportions of studies affected by each of six deficiencies and, in a paired empirical study on public actigraphy cohorts [16], measured how much each of the three most consequential ones costs.

2 Methods

The review follows PRISMA 2020 [13] and was registered on PROSPERO ([TODO: CRD]). The full protocol, search strings, extraction workbook and code are available in the project repository [17].

2.1 Eligibility criteria

Table 1: Inclusion and exclusion criteria.

Element	Inclusion	Exclusion
Population	Human participants, any age or setting, any country	Animal studies; simulation-only work
Condition	Any mental health disorder or state (depressive, anxiety, stress-related, PTSD, psychotic, bipolar, eating, substance use, ADHD, psychological distress)	Neurological conditions without a psychiatric outcome; physical health outcomes
Index model	Supervised ML or DL model developed and/or validated to predict, detect, classify or diagnose a mental health outcome	Descriptive statistics; unsupervised clustering without a predictive target; conventional regression alone; chatbots or interventions without a prediction model
Comparator	Not required	—
Outcomes	At least one quantitative performance metric	No performance metric
Design	Development, validation, or development-with-validation studies; any data source	Reviews, editorials, protocols, abstracts without full text, preprints (analysed as a separate stratum if included)
Timeframe	1 January 2018 to [TODO: date]	Before 2018
Language	English	Non-English

2.2 Information sources and search

PubMed, Scopus, Web of Science Core Collection and IEEE Xplore were searched on [TODO: date] using three concept blocks (machine learning AND mental health AND prediction/detection); the full strings are in Supplementary Table S1 and the repository. PsycINFO [TODO: was / was not] searched [TODO: state reason]. Backward and forward citation chasing was performed for every included study and for the four overlapping reviews [1, 3, 4, 2]. [TODO: Record counts per database from search_log.csv.]

2.3 Selection and data collection

Records were de-duplicated and screened in Rayyan with blinding by two reviewers independently at title/abstract and full-text stages; disagreements were resolved by discussion or a third reviewer. Inter-rater agreement is reported as Cohen's κ . Data were extracted in duplicate into a structured workbook (Supplementary File S2) after a five-study pilot: study characteristics, population, outcome definition and prevalence, data modality, number of candidate predictors, models, validation design, performance metrics, and availability statements.

2.4 Risk of bias and reporting appraisal

Risk of bias and applicability were assessed with PROBAST+AI [10] by two reviewers independently (domains: participants and data, predictors, outcome, analysis). Reporting completeness was scored against the 27 TRIPOD+AI items [9] as fully, partially or not reported, with adherence per study computed as fully reported items over applicable items.

2.5 Outcomes

Primary: PROBAST+AI risk-of-bias judgement per domain and overall; TRIPOD+AI adherence per item and per study. Secondary (pre-specified): (1) any external validation (temporal, geographical or independent cohort; a random hold-out does not qualify); (2) calibration reported by any method (plot, slope/intercept, Brier, ECE, Hosmer–Lemeshow); (3) AUROC reported versus accuracy as the only discrimination metric; (4) leakage-prone evaluation design, coded as present when repeated-measures data were split record-wise, when resampling or feature selection preceded the split, or when a predictor was derived from the outcome; (5) EPV, computed as outcome events divided by candidate predictors and compared with the ≥ 20 (regression) and higher ML expectations discussed in PROBAST+AI [10]; (6) code and data availability with a working link. Studies reporting $\geq 99\%$ accuracy without external validation were examined individually as leakage case studies.

2.6 Synthesis

Proportions with Wilson 95% confidence intervals, overall and stratified by data modality (EHR/registry, questionnaire, wearable/actigraphy, speech/interview, imaging, social-media text), disorder and publication year. Meta-analysis of performance was pre-specified only for clinically homogeneous subsets reporting AUROC with variance; [TODO: state whether performed].

3 Results

3.1 Study selection

[TODO: PRISMA flow: 0 database records, 0 from other sources, 0 duplicates, 0 screened, 0 excluded, 0 full texts assessed, 0 included. Use paper/review/prisma_flow_template.tex.]

3.2 Study characteristics

[TODO: Table: year, country, population, disorder, modality, sample size, prevalence, models. Summarise medians and IQRs.]

3.3 Risk of bias (PROBAST+AI)

[TODO: Stacked bar per domain; proportion high/unclear/low; which signalling questions drive high risk (expected: analysis domain, 4.1 sample size, 4.7 calibration, 4.8 optimism/leakage).]

3.4 Reporting completeness (TRIPOD+AI)

[TODO: Per-item adherence bar chart; median adherence per study; items with lowest adherence (expected: sample size justification, calibration, fairness, code availability, model specification).]

3.5 Secondary outcomes

[TODO: Table with one row per outcome: n/N (% , 95% CI), stratified by modality. Leakage case studies described narratively.]

4 Discussion

4.1 Principal findings

[TODO: Three sentences on the headline proportions.]

4.2 What the deficiencies cost: the companion study

The proportions above describe how widespread each practice is; they do not say how much it matters. The companion study [16] answers that on three public actigraphy cohorts (DEPRESJON, PSYKOSE, HYPERAKTIV; 162 unique participants, 1,818 unique valid day-windows, the 32 healthy controls being shared by the first two cohorts) with the same fixed models, seeds and subject-level bootstrap. Record-wise cross-validation inflated window-level AUROC by 0.065 (95% CI 0.024 to 0.122) to 0.114 (0.060 to 0.181) for depression versus controls, by 0.020 to 0.044 for schizophrenia versus controls, and by 0.130 (0.095 to 0.167) to 0.169 (0.135 to 0.206) for ADHD versus clinical controls, where the honest estimate was at chance (AUROC 0.44 to 0.45): leakage did not merely exaggerate a real signal, it manufactured one. Applying a model trained on one cohort to another without refitting reduced AUROC by 0.13 to 0.21 in the harder direction and collapsed calibration slopes to 0.27 to 0.63 with intercepts of 0.7 to 1.4, i.e. probabilities were both overconfident and shifted, while a deliberate label-shift transfer (depression model applied to ADHD versus clinical controls) fell to chance. Even under honest internal validation, calibration slopes were 0.53 to 0.95 and expected calibration error 0.06 to 0.22, and no published model on these cohorts reports any of these quantities. Subject-level events per candidate predictor were

0.6 to 0.8 in every cohort, one to two orders of magnitude below guidance, although window-level counts make the samples look ten to twenty times larger.

4.3 Comparison with previous reviews

Our findings extend Meehan et al. [3] from regression-based clinical models to the ML and digital-phenotyping literature, and go beyond the descriptive catalogue of Hasan et al. [1] by appraising quality rather than method. Benchmark-level audits of clinical interview corpora [14, 15] reach the same conclusion by a different route: published rankings on DAIC-WOZ are unstable across splits and models learn corpus artefacts.

4.4 Strengths and limitations

Breadth across disorders and modalities; pre-registration; PROBAST+AI and TRIPOD+AI applied in duplicate; a paired empirical companion. Limitations: English-only; [TODO: PycINFO]; appraisal depends on what authors report, so under-reporting and poor conduct cannot always be separated; [TODO: single second screener on a sample, if applicable].

4.5 Implications

Journals in this field should require, as a condition of review, a statement of the unit of splitting for repeated-measures data, a calibration plot or slope, subject-level (not window-level) sample sizes with EPV, and a code link. Authors should treat the TRIPOD+AI checklist as a design document rather than a submission form.

Declarations

Funding: [TODO:]. Conflicts of interest: none declared. Registration: PROSPERO [TODO: CRD]. Data and code: extraction workbook, search log and analysis code at [17].

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